SATA 1D in Brief

Position-dependent correction of the topographic residual ΔC_0 before multichannel azimuth reconstruction

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1. What is corrected

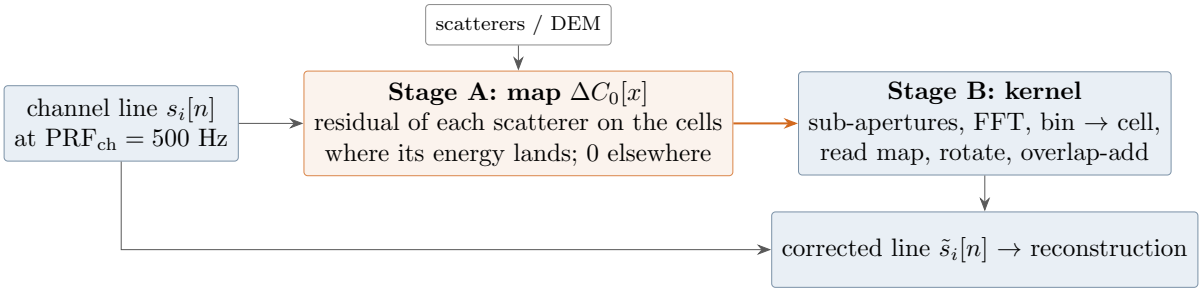
The reconstruction filters of channel i carry the bulk phase $e^{-j2\pi C_{0,i}/\lambda}$ of the reference point p_{ref} on the reference surface. A scatterer at height h sits further out on the iso-range arc, and for a receiver displaced cross-track by $b_{xt,i}$ its bistatic range differs by the *residual*

$$\Delta C_{0,i}(p) = C_{0,i}(p) - C_{0,i}(p_{\mathrm{ref}}) \simeq -\frac{b_{xt,i}\,\Delta h}{r_0 \tan\theta}, \qquad \varphi_i = \frac{2\pi}{\lambda}\Delta C_{0,i}, \tag{1}$$

($\varphi_0 = -78.8°$ for $b_{xt} = 63.7\,\mathrm{m}$ and $h = 240\,\mathrm{m}$.) What degrades the reconstruction is the *spread* of φ_i across channels: the aliased sub-bands are then unmixed incompletely and the remainder appears as azimuth ambiguities ($-13\,\mathrm{dB}$ instead of $-39\,\mathrm{dB}$ for that target). A single multiplication per channel cannot fix it, because φ_i depends on the scatterer and every sample receives many scatterers. SATA selects the phase per scatterer without focusing the line.

2. How SATA 1D works

Principle. In a short sub-aperture the platform hardly moves, so a scatterer is a tone whose Doppler frequency gives its position: $f = (2v/\lambda)\sin\beta$. The Doppler bins of a sub-aperture are positions, and each bin can be rotated by the residual of the position it looks at. Two stages, per channel, before the reconstruction:



Stage B, the kernel (`sata_1d`). Windows of $T_{\mathrm{sub,eff}} = 2\lfloor \delta x\,\mathrm{PRF_{ch}}/2v \rceil = 20$ samples ($\delta x = \sqrt{\lambda r/2} = 309\,\mathrm{m}$), triangular weight, 50 % overlap, zero-padded FFT of $N_{\mathrm{zp}} = 128$ bins. Every bin f is mapped to an azimuth cell through the beam angle, the *ruler*,

$$\beta(f) = \arcsin\frac{\lambda f}{2v}, \qquad \mathrm{posaux}(f) = c_w + r\tan\beta(f)\,\frac{\mathrm{PRF_{ch}}}{v} \;\; (\simeq c_w + 0.81\ \mathrm{cells/Hz}\cdot f), \tag{2}$$

which is the exact inverse of the physics: the bin holding a scatterer's energy always points at the scatterer's own cell x_t, whatever the window. The bin is then multiplied by

$$R(f) = e^{+j2\pi\,\Delta C_0[\mathrm{posaux}(f)]/\lambda}, \tag{3}$$

the conjugate of the residual factor, and the windows are recombined (constant overlap-add). Because the channel is sampled at $\mathrm{PRF}_{\mathrm{ch}}$ while a target sweeps 1281 Hz, its Doppler folds: at the beginning and end of the illumination the peak points at $x_t \mp X$, $X = 405$ cells $= 6.2$ km, one fold.

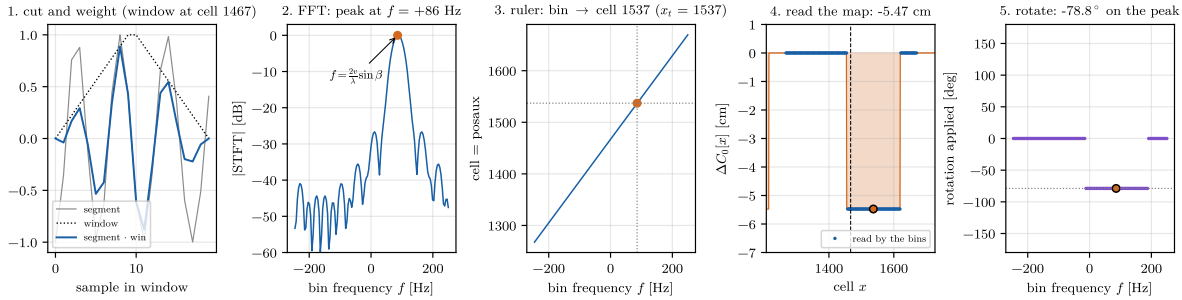
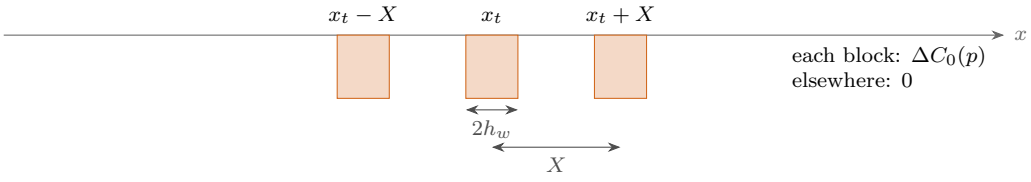


Figure 1: One window of channel 0 (target at $h = 240$ m, cell 1537; window centred 70 cells before it). The spectrum peaks at $+86$ Hz, the ruler turns that into cell 1537, the map holds -5.47 cm there, and the peak is rotated by $-78.8°$: exactly the conjugate of the residual phase. Bins pointing outside the target's block read 0 and are left alone.

Stage A, the map (`build_delta_C0_array`). For each scatterer: (A1) $\Delta C_{0,i}$ from the same range-history fit that builds the filters; (A2) its cell x_t on the channel line, including the $b_{at,i}/2$ shift of the phase centre; (A3) a block $x_t \pm h_w$, $h_w = 82$ cells $= 1.26$ km, the second null of the window response (99.9 % of the energy); (A4) the same block at the fold images $x_t \pm X$; (A5) neighbouring blocks (gap $\leq 2h_w$) are joined and linearly interpolated, overlapping images are split between the nearest components, everything else is zero. The rule the map implements: *a bin is corrected with the residual of the scatterer it shows, and with nothing else.* With two scatterers in one window the spectrum has two peaks, each at its own cell, and each reads its own block.



Whole band or per sub-band. Whole band: one corrected set feeds all output sub-bands. Per sub-band: both stages run once per output band k ($f_k = -750, -250, 250, 750$ Hz), with ΔC_0 fitted over the band's look angles and the kernel labelling its bins with the absolute Doppler in $[f_k - \mathrm{PRF}_{\mathrm{ch}}/2,\, f_k + \mathrm{PRF}_{\mathrm{ch}}/2)$ on the broadside grid ($\beta_{\mathrm{img}} = 0$, report Section 4.2). The previous 1D axis, $f_k +$ fftfreq measured from β_k, was replaced: its two errors cancelled and made the kernel the whole-band one. For iso-range targets ΔC_0 does not depend on the look angle and both variants give the same result.

3. Verification (20 tests, all pass)

	What is checked	Result
T1–T3	zero map = identity; constant map = one global rotation; inject/remove round trip	10^{-16}, 10^{-16}, 0.8 %
T4, T5	residual vs closed form; scatterer cell vs simulated closest approach	0.998–1.000; 0 cells
T6, T7	STFT peak $\to x_t + mX$ (whole band); own-band peak $\to x_t$ (per sub-band)	2.4 cells (<1 bin)
T8, T9	energy of the target read inside the map; corrected channel vs ideal rotation	99.9 %; 2.0 %
T10	one target, $h = 0 \ldots 400$ m: ambiguity vs reference	-39.4 vs -39.1 dB (no SATA: $-23 \ldots -9$)
T12	sign: remove / inject / remove twice	-39.4 / -7.8 / -13.1 dB
T13	two-target map applied to each target alone: reads its own ΔC_0	2.2 %
T11, T14, T17a	multi-target scenes vs the oracle (each target corrected with its own ΔC_0)	within 0.3 dB when resolvable
T15, T16	footprint null (1st–4th), FFT oversampling 1–8	same ambiguity
T17b	two targets vs separation: Doppler-alias collision at $\lvert d - mX\rvert \lesssim 0.6$ km	inherent; outside: oracle
T18, T19	$b_{xt} = 0 \Rightarrow$ identity; Doppler axes (kernel and sub-bands)	5×10^{-11}; exact

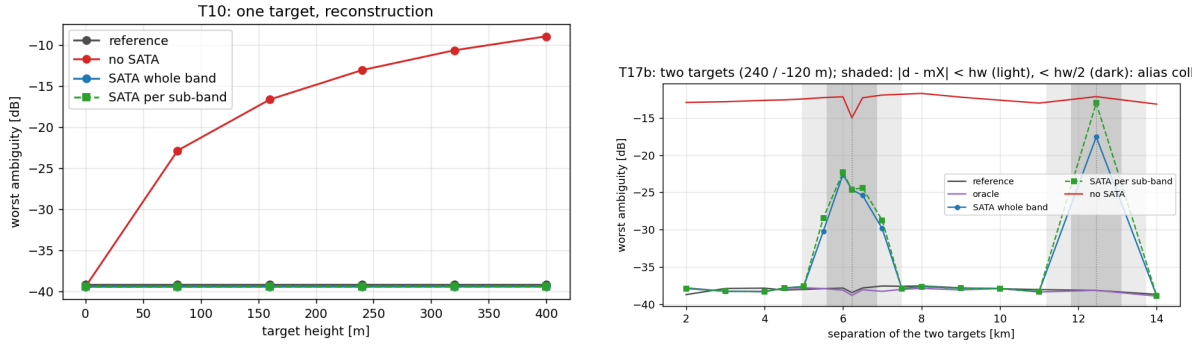


Figure 2: Left: one target, worst ambiguity against height. Right: two targets (240 / -120 m) against their separation; in the shaded bands their Dopplers differ by $m\,\mathrm{PRF}_{\mathrm{ch}}$ and share the same bins, which no per-channel correction can serve.

4. Results and limits

- **One target:** the ambiguities at ± 0.81, ± 1.62 s (-13 dB at 240 m, -9 dB at 400 m) drop to -39.4 dB, the level of the reference, with either variant (Fig. 3).
- **Sweep, 181 scenes of 2–13 targets 200 m apart:** the new map beats the previous one by a median of 3.7 dB (better in 125, equal in 53, worse in 3) and never worsens the uncorrected result. It holds the oracle level as long as neighbouring targets differ by less than about 100 m in height; ramps gain up to 26 dB.
- **Limits of the method:** scatterers closer than $\delta x \simeq 309$ m share one correction; two scatterers a multiple of $X = 6.2$ km apart (within 0.6 km) collide in the same bins; ΔC_1, ΔC_2 are not corrected (3 dB between oracle and reference on dense scenes); targets

displaced in range (topographic ramp) must be corrected on their own range lines, which the 2D processing does.

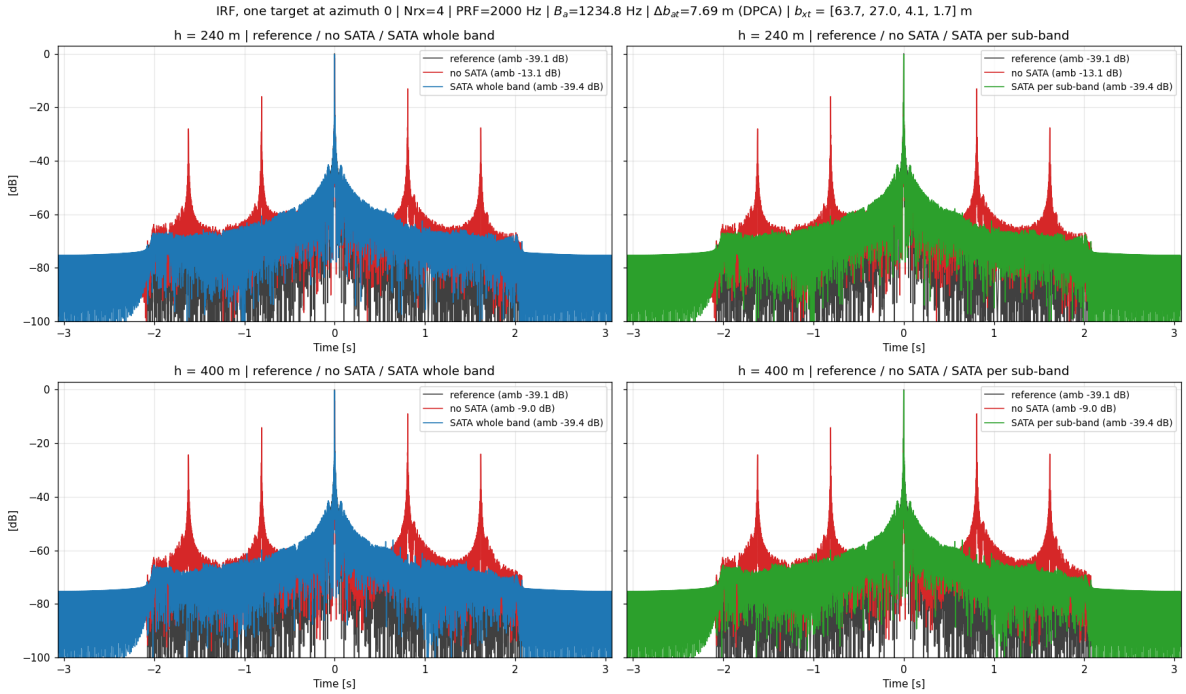


Figure 3: Impulse response of one target at 240 m (top) and 400 m (bottom): reference, no SATA, SATA whole band (left) and per sub-band (right).